DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

UNIVERSITY EXAMINATION 2021/2022 ACADEMIC YEAR

THIRD YEAR SUPPLEMENTARY/SPECIAL EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN ELECTRICAL AND ELECTRONIC ENGINEERING, BACHELOR OF EDUCATION IN TECHNOLOGY (ELECTRICAL AND ELECTRONIC ENGINEERING),
&
BACHELOR OF SCIENCE IN TELECOMMUNICATION AND INFORMATION ENGINEERING

EEE/ETI 3104 DIGITAL ELECTRONICS I

DATE: 14TH APRIL 2022 **TIME: 8:30 - 10:30 AM**

INSTRUCTIONS:
Answer **QUESTION ONE (COMPULSORY)** and **ANY** other **TWO** questions.

QUESTION ONE [COMPULSORY – 30 MARKS]

a) State **THREE** reasons why it is necessary to reduce Boolean expressions before implementing them. **[3 Marks]**

b) Convert each of the following numbers to the indicated base:
   i. 53.8125_{10} to Binary. **[3 Marks]**
   ii. $125 \cdot 36_8$ to Decimal. **[2 Marks]**
   iii. $2CF.A_{16}$ to Binary. **[2 Marks]**
   iv. 1000011001110111_2 to BCD. **[3 Marks]**

c) Prove the following Boolean theorem using a truth table. **[3 Marks]**

$$AB + \overline{A}C + BC = AB + \overline{A}C$$

d) Reduce the following function using Karnaugh Maps:
   i. $F(A,B,C,D) = \sum m(5,7,8,13,14,15)$ in Sum of Products (SOP). **[5 Marks]**
   ii. $F(A,B,C,D) = \prod M(0,1,2,3,4,\ 8,12)$ in Product of Sums (POS). **[5 Marks]**

e) With the aid of examples, explain the following terms as used in Counters?
   i. Modulus of a counter. **[2 Marks]**
   ii. An Up counter. **[2 Marks]**

QUESTION TWO [20 MARKS]

a) A digital circuit has one control line C and three data lines D_1, D_2 and D_3. When the control line is HIGH, the circuit is to detect when one of the data lines has a 1 on it. It is impossible for more than one data line to have a '1' on it. When the control line is LOW, the circuit will output a '0', regardless of what is on the data lines.

   i. Compute a truth table for the above circuit. **[4 Marks]**

   ii. From the truth table in Q4 c(i), derive a Sum-of -Products (SOP) Boolean expression using a Karnaugh- Map. **[4 Marks]**

   iii. Draw a logic diagram for the Boolean expression obtained in Q4 c(ii) above using AND-OR-NOT gates. **[2 Marks]**

b) Distinguish between the following terms as used in digital circuits;

   i. Edge triggering and Level triggering. [**2 Marks]**

   ii. Latches and Flip-flops. **[2 Marks]**

c) State and explain THREE classifications of Counters in Digital systems. **[6 Marks]**

QUESTION THREE [20 MARKS]

a) Draw the circuit diagram of a JK flip flop. **[2 Marks]**

b) Compute a truth table for the JK flip flop in 2 (a) above, where $Q_n$is the present state and Q_{n+1} is the next state. **[12 Marks]**

c) Explain the difference between the following terms:

   i. Synchronous counters and Asynchronous counters. **[2 Marks]**

   ii. Digital systems and Analog systems. **[2 Marks]**

d) Define the term Counter as used in Digital Systems. **[2 Marks]**

QUESTION FOUR [20 MARKS]

In a certain corporation, the four board members A, B, C and D own all stock, which is distributed as follows:

A: 40% B: 30% C: 20% D: 10%

Each member has a percentage vote equal to his holdings and a total vote **greater than 50%** is required to pass a motion. In the boardroom, each member is to have a switch with which to indicate a YES or NO vote. A lamp is to light if the total vote cast is **more than 50%** indicating the motion being voted on is passed. Design an electronic voting system for the corporation and implement the circuit using:

a) NAND gates only. **[10 Marks]**

b) NOR gates only. **[10 Marks]**

QUESTION FIVE [20 MARKS]

a) Draw the block diagram of a **positive-edge triggered** 3-bit Ring Counter and explain its operation by use of timing waveforms. Assume that the initial state of the circuit is $Q_2Q_1Q_0 = 100$

[10 Marks]

b) Explain briefly the following types of clocked flip-flops;

i. Edge-triggered flip flops**.** **[2 Marks]**

ii. Level-driven flip flops. **[2 Marks]**

iii. Master-slave flip flops. **[2 Marks]**

c) Given a Boolean Expression $F(A, B, C) = AB\left(\overline{\overline{A} + BC}\right)$;

i. Simplify the above Boolean expression using Boolean algebra. **[3 Marks]**

ii. Draw the logic diagram of the simplified expressed obtained in 5 c (i). **[1 Mark]**